A corollary of the ergodic theorem

Terence Tao · 17 July 2018

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Let (X, T, μ) be a measure-preserving system – a probability space (X, μ) equipped with a measure-preserving translation T (which for simplicity of discussion we shall assume to be invertible). We will informally think of two points x, y in this space as being "close" if $y = T^n x$ for some n that is not too large; this allows one to distinguish between "local" structure at a point x (in which one only looks at nearby points $T^n x$ for moderately large n) and "global" structure (in which one looks at the entire space X). The local/global distinction is also known as the time-averaged/space-averaged distinction in ergodic theory.

A measure-preserving system is said to be ergodic if all the invariant sets are either zero measure or full measure. An equivalent form of this statement is that any measurable function $f : X \to \mathbf{R}$ which is *locally essentially constant* in the sense that $f(Tx) = f(x)$ for μ-almost every x, is necessarily *globally essentially constant* in the sense that there is a constant c such that $f(x) = c$ for μ-almost every x. A basic consequence of ergodicity is the mean ergodic theorem: if $f \in L^2(X, \mu)$, then the averages $x \mapsto \frac{1}{N} \sum_{n=1}^N f(T^n x)$ converge in L^2 norm to the mean $\int_X f \, d\mu$. (The mean ergodic theorem also applies to other L^p spaces with $1 < p < \infty$, though it is usually proven first in the Hilbert space L^2.) Informally: in ergodic systems, time averages are asymptotically equal to space averages. Specialising to the case of indicator functions, this implies in particular that $\frac{1}{N} \sum_{n=1}^N \mu(E \cap T^n E)$ converges to $\mu(E)^2$ for any measurable set E.

In this short note I would like to use the mean ergodic theorem to show that ergodic systems also have the property that "somewhat locally constant" functions are necessarily "somewhat globally constant"; this is not a deep observation, and probably already in the literature, but I found it a cute statement that I had not previously seen. More precisely:

> **Corollary 1** Let (X, T, μ) be an ergodic measure-preserving system, and let $f : X \to \mathbf{R}$ be measurable. Suppose that
>
> $$\limsup_{N \to \infty} \frac{1}{N} \sum_{n=1}^N \mu(\{x \in X : f(T^n x) = f(x)\}) \geq \delta \tag{1}$$
>
> for some $0 \leq \delta \leq 1$. Then there exists a constant c such that $f(x) = c$ for x in a set of measure at least δ.

Informally: if f is locally constant on pairs $x, T^n x$ at least δ of the time, then f is globally constant at least δ of the time. Of course the claim fails if the ergodicity hypothesis is dropped, as one can simply take f to be an invariant function that is not essentially constant, such as the indicator function of an invariant set of intermediate measure. This corollary can be viewed as a manifestation of the general principle that ergodic systems have the same "global" (or "space-averaged") behaviour as "local" (or "time-averaged") behaviour, in contrast to non-ergodic systems in which local properties do not automatically transfer over to their global counterparts.

Proof: By composing f with (say) the arctangent function, we may assume without loss of generality that f is bounded. Let $k > 0$, and partition X as $\bigcup_{m \in \mathbf{Z}} E_{m,k}$, where $E_{m,k}$ is the level set

$$E_{m,k} := \{x \in X : m 2^{-k} \leq f(x) < (m+1) 2^{-k}\}.$$

For each k, only finitely many of the $E_{m,k}$ are non-empty. By (1), one has

$$\limsup_{N\to\infty} \sum_m \frac{1}{N} \sum_{n=1}^N \mu(E_{m,k} \cap T^n E_{m,k}) \geq \delta.$$

Using the ergodic theorem, we conclude that

$$\sum_m \mu(E_{m,k})^2 \geq \delta.$$

On the other hand, $\sum_m \mu(E_{m,k}) = 1$. Thus there exists m_k such that $\mu(E_{m_k,k}) \geq \delta$, thus

$$\mu(\{x \in X : m_k 2^{-k} \leq f(x) < (m_k+1)2^{-k}\}) \geq \delta.$$

By the Bolzano-Weierstrass theorem, we may pass to a subsequence where $m_k 2^{-k}$ converges to a limit c, then we have

$$\mu(\{x \in X : c - 2^{-k} \leq f(x) \leq c + 2^{-k}\}) \geq \delta$$

for infinitely many k, and hence

$$\mu(\{x \in X : f(x) = c\}) \geq \delta.$$

The claim follows. □

Comments (11)

 **burrobert** · 17 July 2018

Thanks Terry. Short and sweet. Nice result and demonstration of some basic analysis. The $ \varepsilon$ and 2^{-k} correspond to the same object. I think $c - 2^{-k}$ in the second last equation could be replaced by c.
[Corrected, thanks – T.]

 **Aritra Bhattacharya** · 18 July 2018

In the first line of proof, do you mean $tan(f(x))$ will be bounded?
[Corrected to arctangent – T.]

 **Maths student** · 20 July 2018

Dear Prof. Tao,
I think only finitely many of the sets $E_{n,\epsilon}$ have nonzero measure; we may put in points with zero measure without changing anything.
Did you ever intend to put all your ergodic lectures somehow in one file, so that it's easily printable??? I'd be VERY interested.

>  **Maths student** · 20 July 2018
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> I'd be able to extract the code by a clever S&R program if the math weren't images, hence the need...

>  **Maths student** · 20 July 2018
>
> I take the correction back, we do have boundedness.

> >  **Maths student** · 21 July 2018
> >
> > But still, it's the $E_{m,\epsilon}$ and not the $E_{m,k}$.
> > *[Corrected, thanks – T.]*

 **Maths student** · 20 July 2018

Ah it's in the Poincaré legacies book.

 **Anonymous** · 22 July 2018

It seems that ergodicity (similarly to analyticity or even convexity) has the potential to "propagate" certain local properties into global ones.

 **Rogier Brussee** · 24 August 2018

I think you want to compose with the arctangent rather than the tangent function.
[Huh, I thought I had fixed that already. In any case, fixed again – T.]

 **Fourier uniformity of bounded multiplicative functions in short intervals on average | What's new** · 5 December 2018

[...] that ths quantity is essentially constant in (cf. the application of the ergodic theorem in this previous blog post), thus we now [...]

 **majdoddin** · 22 June 2019

It seems to me that Corollary 1 can be stated more generally:
Let (X, T, μ) be an ergodic measure-preserving system, and let $f : X \to \mathbf{R}$ be measurable. Suppose that

$$\limsup_{N\to\infty} \frac{1}{N}\sum_{n=1}^{N} \mu(\{x \in X : |f(T^n x) - f(x)| \leq \epsilon\}) \geq \delta \qquad (1)$$

for some $0 \leq \epsilon, \delta \leq 1$. Then there exists a set $E \subseteq X$ of measure at least δ such that $\max_{x\in E} f(x) - \min_{x\in E} f(x) \leq \epsilon$.

With Corollary 1 as a special case, and a very similar proof.